Video Recording Script: Harvest Ledger

About 13 to 15 minutes at a calm pace. Orange italics are actions on screen; read the black text aloud. For each chart: what it shows, how to read it, what it tells us. Where a number depends on the filters, read it from the screen.

0. Intro and audience

ON SCREEN: Open index.html on the Scorecard.

Good day, Professor. We are Mendez and Subala. Our final project is an interactive dashboard on India crop production.

As you explained, a dashboard should serve one specific audience and show only what that audience needs. Ours is built for an agriculture policy planner, someone at a state or national agriculture department who decides where to focus support.

So the dashboard answers four planner questions: how dependent is our supply on one crop, which states carry it, is production growing, and where should land or support be shifted.

1. The data

ON SCREEN: Stay on the Scorecard and point to the header.

The data is from India's Ministry of Agriculture, through Kaggle. Each row says how much of a crop was planted and produced in a district, in a year and season, from 1997 to 2015.

There are 246,091 rows, 33 states, 646 districts and 124 crops. Production is mostly in tonnes, but coconut is counted in nuts, so totals describe recorded volume, not weight or value. Please keep that in mind.

We removed 3,730 rows with missing production, leaving 242,361. Raw has more rows than cleaned because cleaning only removes rows.

2. Filters

ON SCREEN: Click the Cleaned / Raw switch, then point along the filter bar.

These filters apply to the planner pages: Cleaned or Raw data, year range, state, season and crop. Show Top controls how many items the ranked charts display, and Exclude coconut is optional. Every chart has a Chart or Table switch and a written explanation.

PAGE 1: SCORECARD

Scorecard: the six KPIs

ON SCREEN: Point to each KPI card in turn, then to the rating note and key takeaways.

A planner wants answers at a glance, so the top of the page is six KPIs, each with a status label.

Total production is 141.18 billion in recorded volume. Annual growth is plus 2.5 percent, rated Growing; we skip the sparse first and last years when computing it.

Top-crop share is 92.1 percent, from coconut, rated High dependence. Top-3 state share is 90.2 percent, from Kerala, Andhra Pradesh and Tamil Nadu, rated Highly concentrated.

Crops for 80 percent of output is just one crop out of 124, rated Very narrow. Output stability is 0.26, rated Variable, where lower means steadier.

The ratings are our rules of thumb, listed under the cards. The key takeaways below turn them into crop risk, regional risk, momentum and reliability.

Chart: Production through time (area line chart)

Shows: Total recorded production for each crop year.

How to read it: Each point adds up every record in that year. A rising line means more recorded output.

What it tells us: Production peaked in 2011 at about 14.3 billion, around 1.9 times the average year. The drops at the very start and end are incomplete reporting, not a real collapse, and the page says so.

Charts: Top 10 crops and Top 10 states (horizontal bars)

Shows: The crops and states with the highest total recorded production.

How to read it: Longer bar means more recorded volume; the label is the exact total.

What it tells us: Coconut is about 129.98 billion, 23.5 times the next crop, sugarcane. Kerala is 69.3 percent, ahead of Andhra Pradesh at 12.3 percent. States rank high because they grow a high-volume crop like coconut, not only because of land.

Exclude coconut

ON SCREEN: Turn Exclude coconut ON, watch the KPIs change, then OFF.

Because coconut dominates, we added an optional switch. With it on, the dependence and concentration KPIs change, which shows how much one crop drives the picture. It only changes the display, not the data.

PAGE 2: TRENDS

Chart: Total production by year (line chart)

Shows: Production for every year; the Table view adds year-over-year percentages.

How to read it: Read the overall direction first, then look for years that break the pattern.

What it tells us: Read the numbers shown on screen: how many years rose and fell, and the compound growth rate between 1998 and 2014. Edge years are not trusted, as on the Scorecard.

Chart: Yield and area (dual-axis line chart)

Shows: Yield, which is production per hectare, on the left axis, and cultivated area on the right.

How to read it: If production rises faster than area, the gain comes from better yield rather than more land.

What it tells us: It tells the planner whether growth comes from more land or from higher productivity. Read the percentages from the screen. Yield here mixes crops with different units, so it is a broad indicator.

PAGE 3: CROPS

Chart: Top crops by production (bar chart)

Shows: Ranking of crops by total production, reacting to the Show Top setting.

How to read it: Longer bar, more volume.

What it tells us: It shows how few crops carry the volume. The page states how many crops are needed to reach 80 percent of production.

Chart: Concentration, Pareto (line chart)

Shows: The running percentage of production as crops are added from largest to smallest.

How to read it: A line that shoots up and flattens means a few crops dominate.

What it tells us: The first crop alone is more than half of everything, so supply risk sits in one crop.

Chart: Top crops by yield (bar chart)

Shows: Production per hectare for each crop.

How to read it: Longer bar means more output from each unit of land.

What it tells us: Yield removes the effect of how much land a crop occupies, so high volume and high yield are different things. Read the leading crops from the screen, and remember units differ between crops.

PAGE 4: REGIONS

Chart: Top states (bar chart)

Shows: States ranked by total production; the Table view adds share, area, yield and record count.

How to read it: Longer bar, bigger contribution.

What it tells us: A few states supply 80 percent of production. The table shows whether a high rank comes from land, productivity or simply more reporting.

Chart: Output stability (bar chart of variation)

Shows: How much each state's yearly production varies relative to its average, using the coefficient of variation.

How to read it: A shorter bar means a more dependable state. Below about 0.3 is steady; above 1 swings wildly.

What it tells us: It tells the planner which states can be relied on. High variation can also come from reporting gaps, and the page says so.

PAGE 5: SEASONS

Chart: Production by season (bar chart)

Shows: Total production in each of the six seasons.

How to read it: Each bar is a season; the table gives shares.

What it tells us: Whole Year is the largest because of perennial crops like coconut and sugarcane, not because it is a main cropping season.

Chart: Which crops fill each season (heatmap)

Shows: The percent of each season's production supplied by each top crop.

How to read it: Each row is a season, each column a crop; darker means a bigger share.

What it tells us: It shows which crops the supply in each season depends on. The top crops shown may not add to 100 percent.

PAGE 6: PREDICT YIELD (advanced model)

Why this model

ON SCREEN: Click Predict yield and point to the five KPI cards.

To go beyond describing the past, we added one advanced model. We chose predictive and prescriptive analytics because a planner needs to know what yield to expect and where to act. NLP does not fit because the data has no free text.

We predict crop yield with a Gradient Boosting model using state, crop, season, year and area. We trained it on 1997 to 2010, 190,535 records, and tested it on 2011 to 2014, 51,265 records it never saw.

The accuracy, R squared, is 0.79 against 0.73 for a simple baseline that guesses each crop's average. The average error is 39 percent lower.

Chart: Model vs baseline (bar chart)

Shows: The average error, MAE, and the root mean squared error, RMSE, for the baseline and for the model.

How to read it: Shorter bars are better.

What it tells us: The model beats the baseline on both. The baseline is already strong because most of the variation in yield comes from knowing which crop it is, and the model's extra value comes from the state.

Chart: Actual vs predicted yield (scatter plot)

Shows: A sample of test records from 2011 to 2014, actual yield against the model's prediction, on a log scale.

How to read it: A dot on the dashed line is a perfect prediction; the further away, the bigger the miss.

What it tells us: The cloud follows the line closely, so the model tracks the level of yield well, but weather and inputs are not in the data, so it cannot explain a bad or bumper year.

Chart: What drives yield (bar chart)

Shows: How much accuracy drops when each feature is shuffled.

How to read it: A longer bar means the model relies on that feature more.

What it tells us: Crop matters most, then state. Season and area add little, and crop year adds nothing, which means there is no steady yield trend across the years.

Chart: Typical error by crop (bar chart)

Shows: The median percentage gap between predicted and actual yield for the main crops.

How to read it: A shorter bar means better predictions.

What it tells us: Sugarcane is best at about 17 percent, and cotton is hardest at about 30 percent. So the predictions are for ranking and comparing, not for exact forecasts.

PAGE 7: RECOMMEND (prescriptive)

How the recommendation works

ON SCREEN: Click Recommend; point to the warning banner and the KPI cards.

The recommendation uses the model's predicted yields. For each state, season and crop we build a yield index: predicted yield divided by the typical yield of the same crop across states. Above 1 means the state is expected to do better than average, and an index lets crops with different units be compared.

Then a linear program, per state and season, shifts land towards crops with a high index and away from crops with a low index. Total area stays the same, no more than 10 percent of the land moves, and each crop changes by at most 25 percent. Only clear winners and laggards are moved; the rest are held.

Please note the banner: there are no price or cost data, so better means a higher predicted yield, not higher profit. This is decision support, not an instruction.

Across 25 states there are 102 expand recommendations, 112 reduce and 553 hold.

Chart: Where reallocation helps most (bar chart)

Shows: For each state, the estimated gain in area-weighted yield index from the recommended shifts. This is shown when All states is selected.

How to read it: A longer bar means more room to improve.

What it tells us: Kerala has the most at about 11.6 percent, then Meghalaya and Arunachal Pradesh. A bigger gain means a bigger mismatch between current crops and where the state yields best, not that the state performs badly.

Chart and table: Crop plan for one state (use the State selector)

Shows: For the chosen state, which crops to expand or reduce, and by how much.

How to read it: Green bars add area, rust bars trim it. The table shows area now, recommended area, yield index and yield variation.

What it tells us: Select Gujarat: it gains about 3.4 percent by expanding castor seed, banana, onion, potato, sesamum, garlic and arhar, and reducing Kharif rice and maize and Whole Year sugarcane. Before expanding a crop, check its yield variation; above about 0.3 means it has swung a lot.

PAGE 8: DATA NOTES

Data notes

ON SCREEN: Click Data notes.

Data notes shows raw and cleaned row counts, a completeness chart where only Production has gaps, the field definitions, how each measure is calculated, and the cleaning steps and limitations. It lets the planner see how far the numbers can be trusted.

Limitations and closing

ON SCREEN: Return to the Scorecard.

There are limits. Units are mixed, the first and last years are incomplete, we removed rows with missing production, the model has no weather or price data, and the model results do not change with the filters.

In conclusion, India's recorded crop production is extremely concentrated, in coconut and in Kerala. For a planner, that is the main supply risk. The dashboard turns the descriptive findings into six KPIs, predicts yield, and recommends where to shift land. Thank you, Professor.

Tips for recording

- Open the dashboard and let it finish loading before you press record. Use a full-screen browser at 100% zoom and move the mouse slowly.
- Pause 1 to 2 seconds after each page change so the chart is visible.
- If a number on screen differs from this script, read the screen: filters change the numbers on the first six pages.

- If you are short on time, keep: the six KPIs, Production through time, Top crops, Mean error vs baseline, What drives yield, and the Gujarat crop plan.
- Likely questions: why these KPIs (they answer the planner's decisions); why yield not production (planners can influence yield); why a time-based split (to test on years the model never saw); why are rating cut-offs fixed (rules of thumb, adjustable).